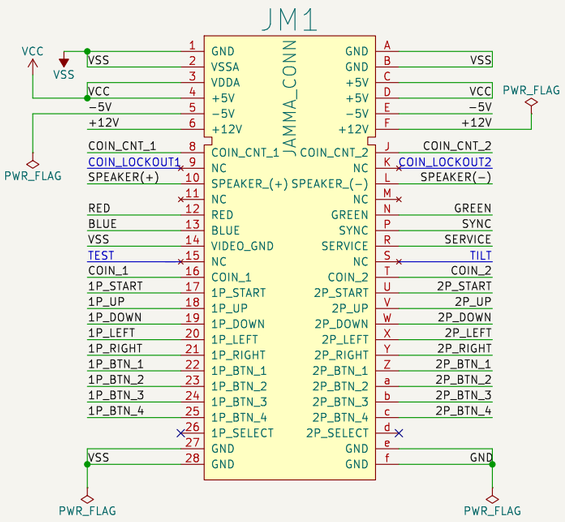
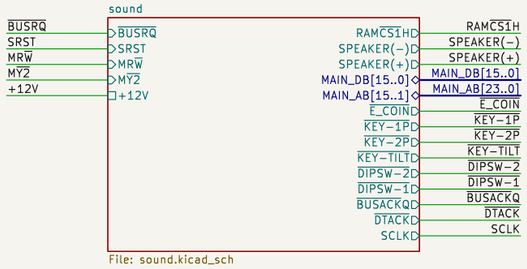
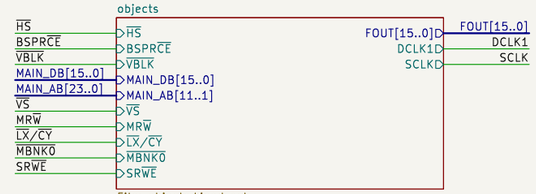
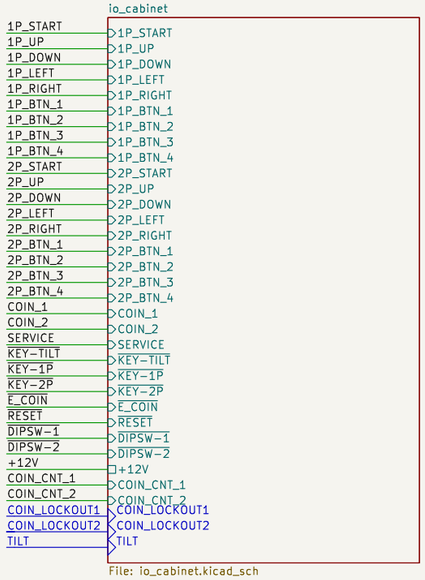
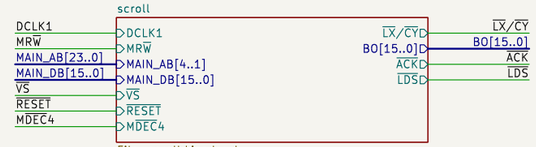






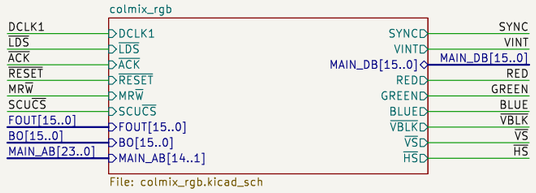










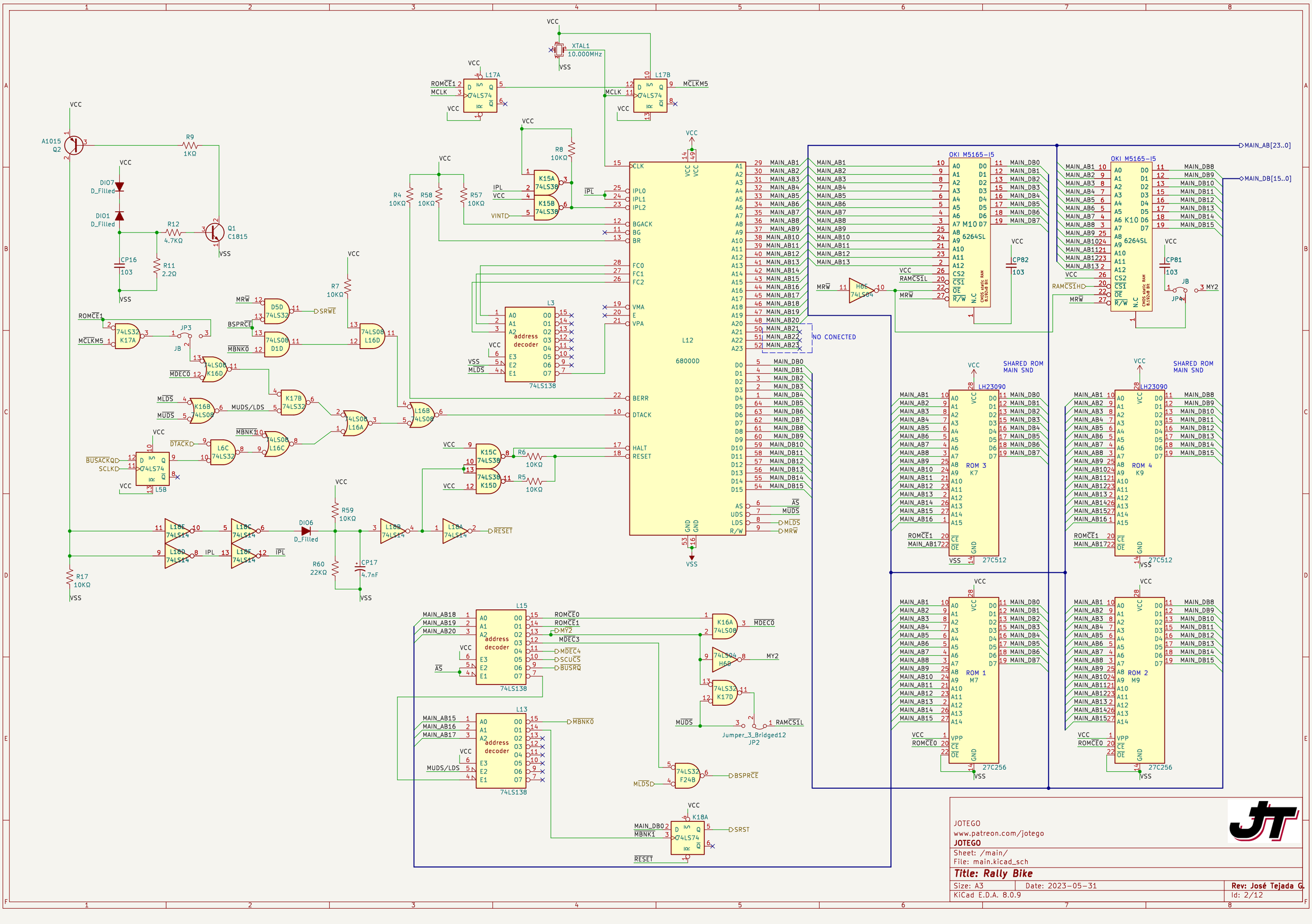


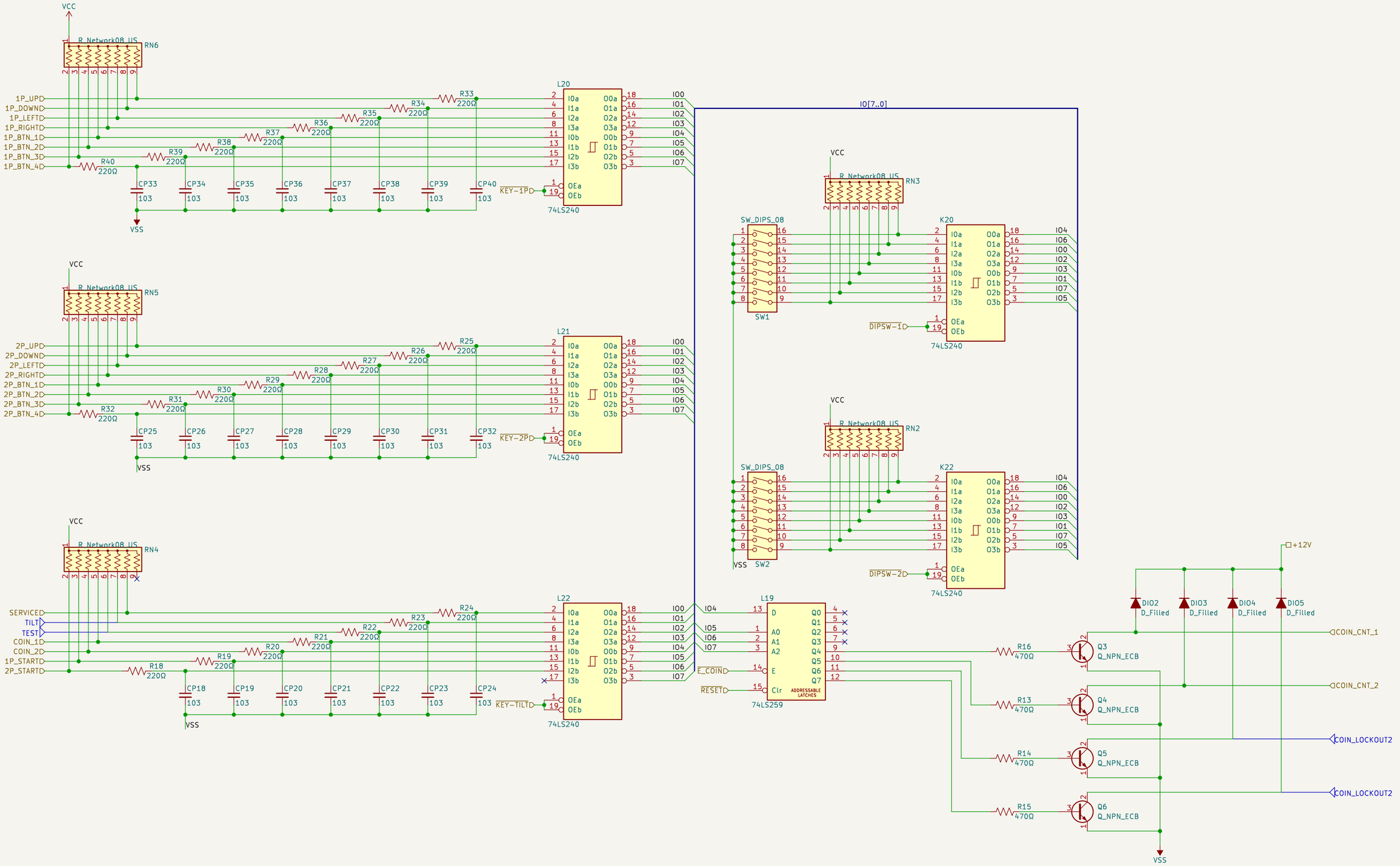
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Sheet: /
File: rbike.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023−05−31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 1/12


XTAL1
10.000MHz
VCC
VSS
L17A
L17B
74LS74
ROMCE1
MCLK
MCLKM5
L12
68000D
CLK
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
MAIN_AB[23..0]
MAIN_DB[15..0]
NO CONECTED
OKI M5165-I5
6264SL
CP82
CP81
JB
JP4
MY2
RAMCS1H
RAMCS1L
SHARED ROM
MAIN SND
LH23090
ROM 3
K7
ROM 4
K9
ROM 1
M7
ROM 2
M9
27C512
27C256
L3
L15
L13
74LS138
address decoder
L5B
L6C
L16A
L16B
L16C
L16D
L16E
L16F
K15A
K15B
K15C
K15D
K16A
K16B
K17A
K17B
K18A
D5D
D1D
F24B
H6B
M6
IPL
VINT
MRW
MDS
MUDS
MBNK0
MBNK1
MDEC0
MDEC3
MDEC4
SCUCS
BUSRQ
SRWE
BSPRCE
MUDS/LDS
BUSACKQ
SCLKD
DTACK
SRST
Jumper_3_Bridged12
JP2
A1015
Q2
Q1
C1815
DIO7
DIO1
DIO6
D_Filled
R4
R5
R6
R7
R8
R9
R11
R12
R17
R57
R58
R59
R60
CP16
CP17
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Sheet: /main/
File: main.kicad_sch
Title: Rally Bike
Size: A3
Date: 2023-05-31
Rev: José Tejada G.
KiCad E.D.A. 8.0.9
Id: 2/12

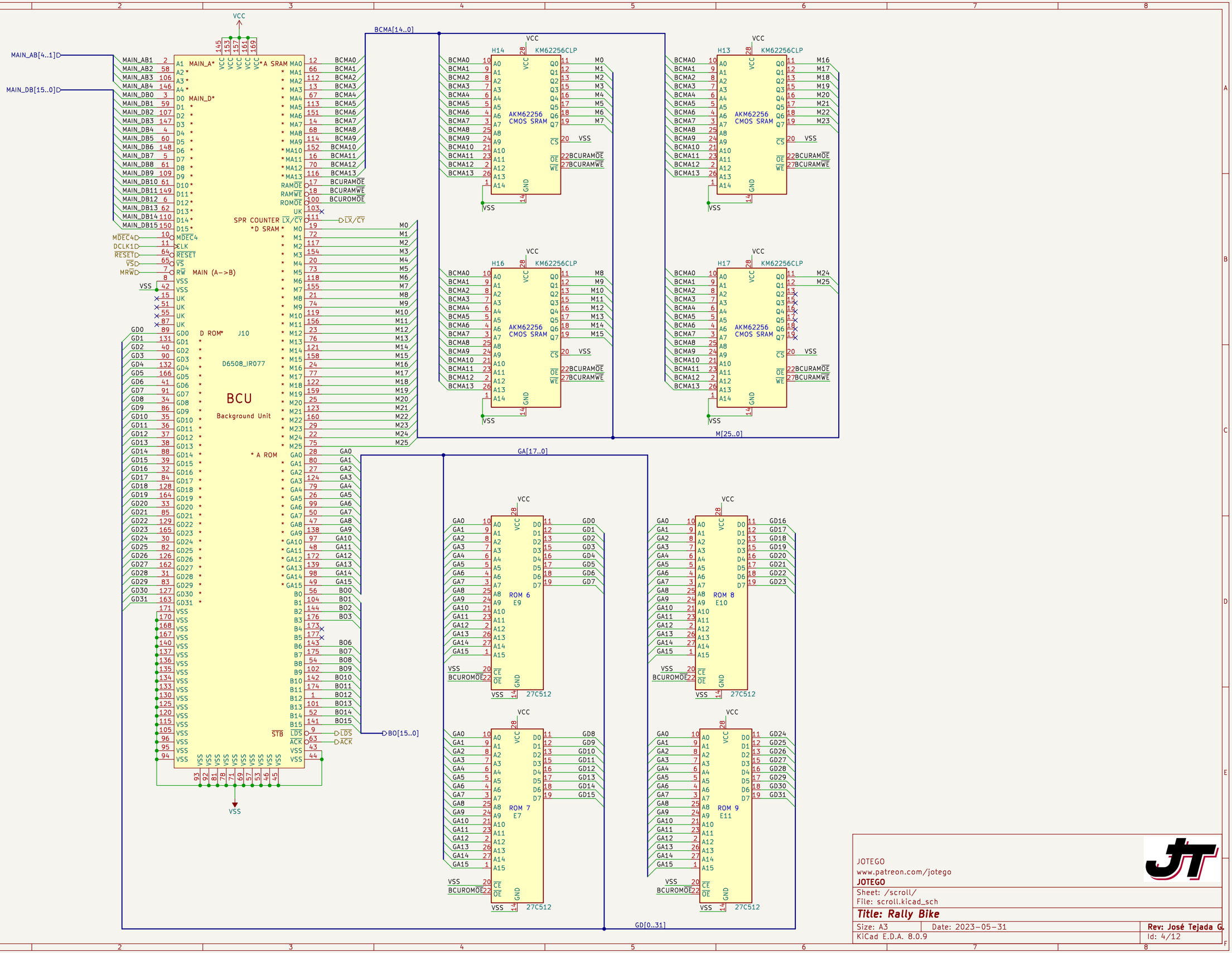


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Sheet: /io_cabinet/
File: io_cabinet.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 3/12


BCMA[14..0]
MAIN_AB[4..1]
MAIN_DB[15..0]
BCU
Background Unit
D6508_IR077
MAIN (A->B)
D ROM*
J10
* A ROM
* D SRAM
* A SRAM
SPR COUNTER
BO[15..0]
GA[17..0]
GD[0..31]
M[25..0]
H14
H13
H16
H17
AKM62256 CMOS SRAM
KM62256CLP
ROM 6 E9
ROM 7 E7
ROM 8 E10
ROM 9 E11
27C512
VCC
VSS
JOTEGO
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Sheet: /scroll/
File: scroll.kicad_sch
Title: Rally Bike
Size: A3
Date: 2023-05-31
Rev: José Tejada G.
KiCad E.D.A. 8.0.9
Id: 4/12

obj_memories

MAIN_DB[15..0]
MAIN_AB[11..1]
VBLK
COUNTQ2
MRW
BSPRCE
SPR'OE
SPRAMOE
MBNK0
DCLK2
LX/CY
V5
SRWE
RW
COUNTQ4
LD
CLR
Cp3
Cp1
Cp2
Cp4
Cp5
SCM[10..0]
SCX[2..0]
COUNT[3..0]
HSPRD'[15..0]
SPRD'[15..0]
VBLK
SPRDB15

File: obj_memories.kicad_sch

obj_pxl

SRCLK
SPROMA0
COUNT[3..0]
SPRWEX
SPRWEY
FX
FY
LATCH1
SPRD'[10..0]
HSPRD'[15..0]
FY[15..0]
FX[11..0]
SHIFTR

File: obj_pxl.kicad_sch

Obj_fms

SHIFTR
FY
VBLK
LX/CY
DCLK2
WE
RW
SPRCET
SRCLK
CLR
WEY
COUNTQ2
SCDPE
COUNTQ4
SPROMA0
SPRAMOE
SPRCEP
LATCH1
SPR'OE
LD
Cp1
Cp2
Cp3
Cp4
Cp5
SCLK
SRCLK
DCLK1

File: Obj_fms.kicad_sch

SCM[10..0]
SCX[2..0]

C19 — 74LS174 — hex flip flop

SCM1 SCM2 SCX1 SCX0 SCX2 SCM0 FX VCC
D0 D1 D2 D3 D4 D5 Cp Mr
Q0 Q1 Q2 Q3 Q4 Q5
SCM1' SCM2' SCX1' SCX0' SCX2' SCM0'

C19 — 74LS174 — hex flip flop

SCM4 SCM3 SCM5 FX VCC
SCM4' SCM3' SCM5'

COUNT DIRECTION CONTROL

SPRD'[15..0]

B19 — 74LS169

SPRD'2 SPRD'3 SPRD'4 SPRD'5 SCDPE SPRD'15 SRCLK
P0 P1 P2 P3 PE U/D CEP CET CP TC
Q0 Q1 Q2 Q3
SPRDQ2 SPRDQ3 SPRDQ4 SPRDQ5

SPRCET SPRCEP — C6A 74LS08

C19 — 74LS169

SPRD'6 SPRD'7 SPRD'13 SPRD'14 SCDPE SPRD'15 SRCLK
SPRDQ6 SPRDQ7 SPRDQ13 SPRDQ14

D19 — 74LS169

SPRDB15 SCDPE SPRD'15 SRCLK
SPRQ15

SRCLK SCDPE — D7A 74LS32

SPRD'15 VCC — E24B 74LS74

DCLK1 — 74LS04 D2A

VCC — H24A 74LS74 — HS

VCC — E24A 74LS74 — FY FX

DCLK2 — F24B 74LS32
DCLK2 — F24C 74LS32

WEY SPRWEY — H21A 74LS08
SHIFTR SRCLK WE — 74LS20 H22B
H21D 74LS08 — SPRWEX

XY[17..0]

B20 A20 C20 A21 B21 C21 — Quad Mux — 74LS157

SCM1' SCM2' SCM3' SCM4' SCX0' SCX1' SCX2' SCM0' SCM5'
SPRDQ6 SPRDQ7 SPRDQ13 SPRDQ14 SPRDQ2 SPRDQ3 SPRDQ4 SPRDQ5 SPRQ15
FY FX VSS
IOa IOb IOc IOd I1a I1b I1c I1d Za Zb Zc Zd
SELECT_A-B STROBE_G
XY0 XY1 XY2 XY3 XY4 XY5 XY6 XY7 XY8 XY9 XY10 XY11 XY12 XY13 XY14 XY15 XY16 XY17

AM2148-55PC — 1024X4 Static RAM

A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 CS WE D1 D2 D3 D4
VSS SPRWEY SPRWEX

B22 A24 A22 A23 B23 B24

FY[15..0]
FX[11..0]
FY0 FY1 FY2 FY3 FY4 FY5 FY6 FY7 FY8 FY9 FY14 FY15
FX0 FX1 FX2 FX3 FX4 FX5 FX6 FX7 FX8 FX9 FX10 FX11

DEBERIA ESTAR EN HOJA DE SCU?

74LS373 — LATCHES AND FLIP FLOPS

DCLK2 FY — D22
DCLK2 FX — C24
DCLK2 FY — C22
DCLK2 FX — D24
D0 D1 D2 D3 D4 D5 D6 D7 LE OE
O0 O1 O2 O3 O4 O5 O6 O7

FO[15..0]
FO0 FO1 FO2 FO3 FO4 FO5 FO6 FO7 FO8 FO9 FO14 FO15

74LS374 — 3-state flip flops

DCLK2 — H20
DCLK2 — H19
Cp OE

SCU BUS (FRONT)
FOUT[15..0]
FOUT0 FOUT1 FOUT2 FOUT3 FOUT4 FOUT5 FOUT6 FOUT7 FOUT8 FOUT9 FOUT14 FOUT15

VSS

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Sheet: /objects/
File: objects.kicad_sch

Title: *Rally Bike*

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 5/12

P POR CONSULTAR

SPRITES COUNTER

SYNCHRONOUS 4-BIT UP/DOWN COUNTER

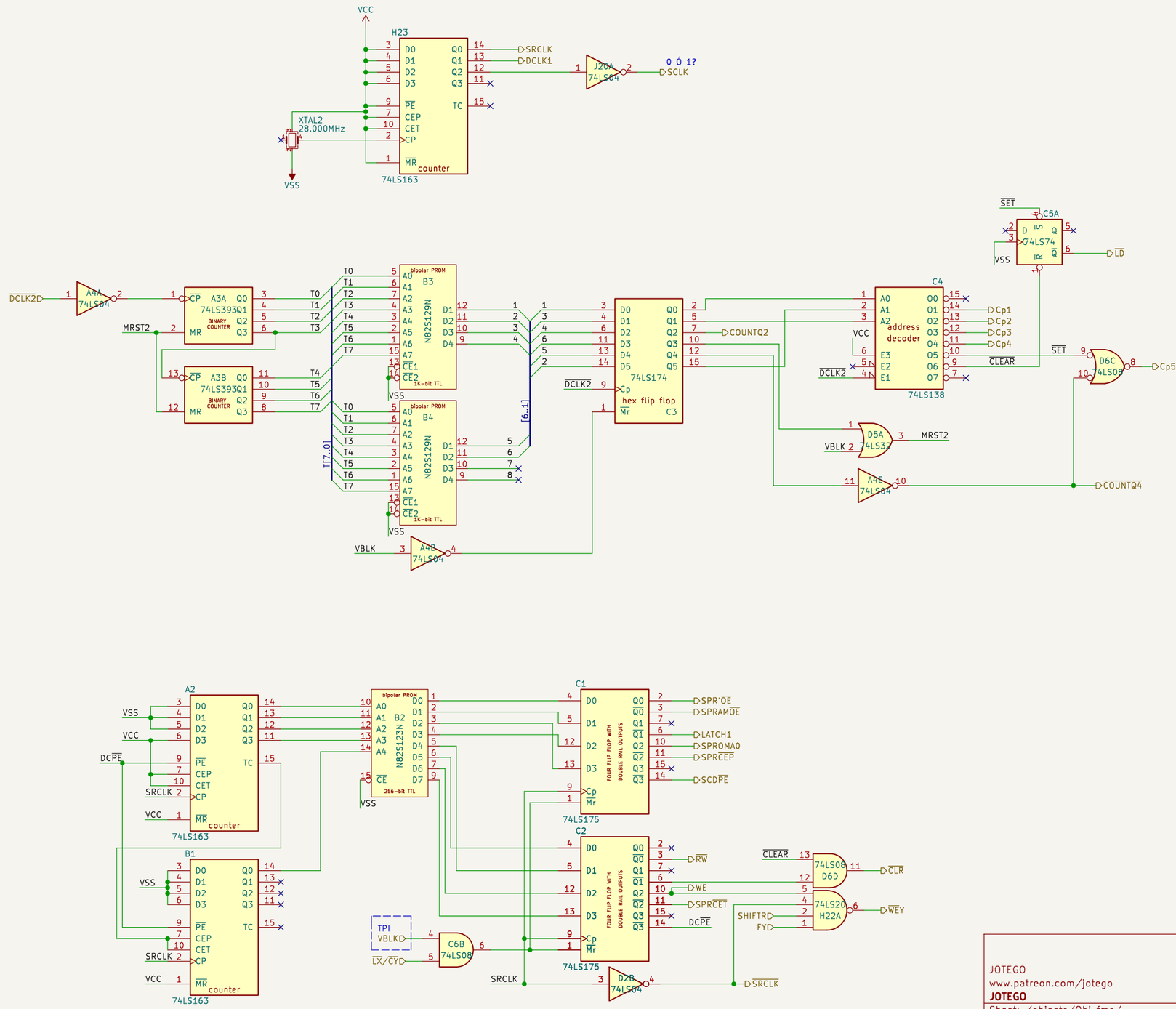
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File: obj_memories.kicad_sch

Title: Rally Bike

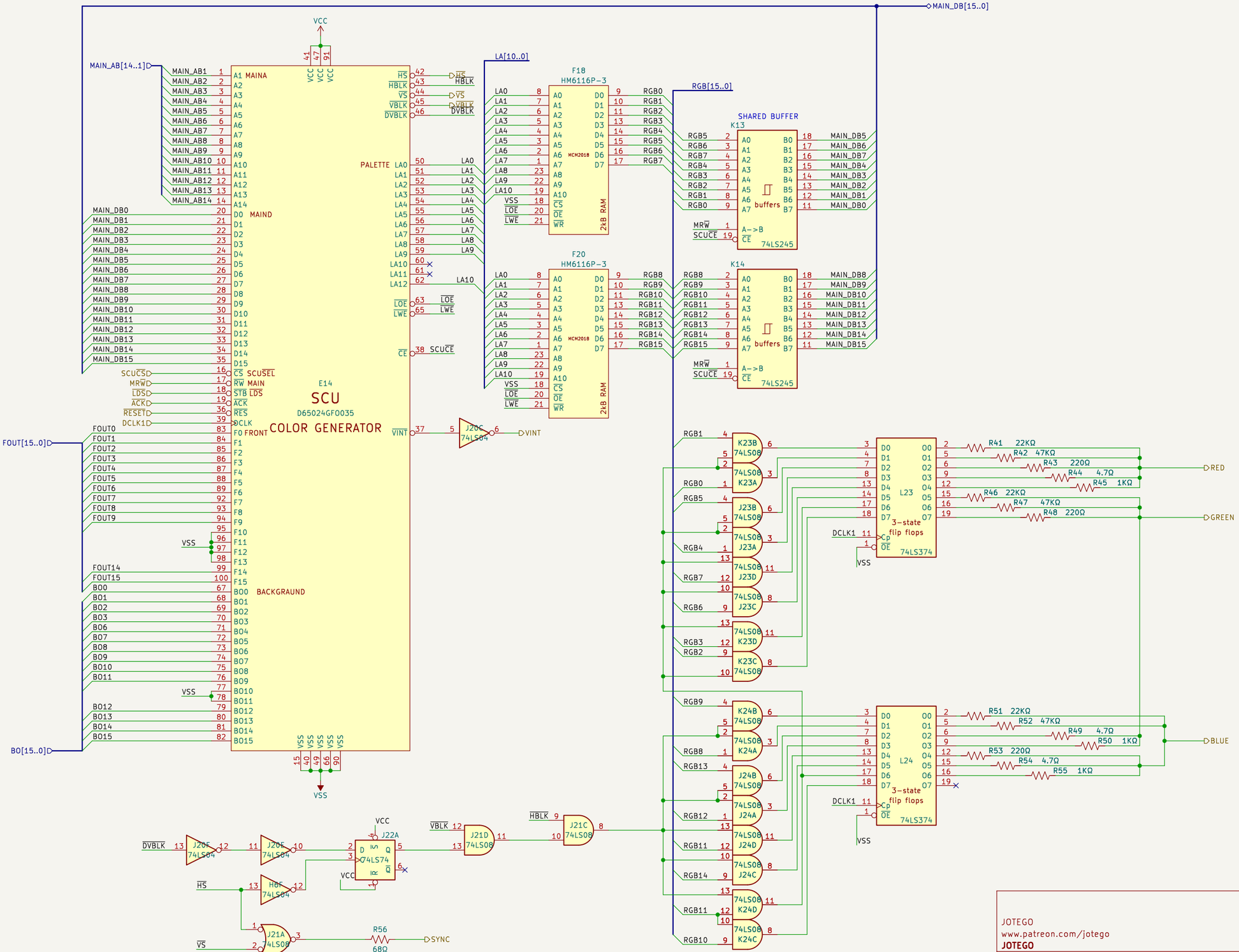
Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 6/12

COUNT[3..0]
SPROMA0
SPROMA0
SPROMA0
SPROMA0
HSPRD'[15..0]
SPRD'[10..0]
SR[3..0]
SHIFT REGISTERS
SHIFTR
FY[15..0]
FX[11..0]
LATCH1
SRCLK
SRCLKD
SPRWEY
SPRWEX
VCC
VSS
GND
F22
F21
E23
E22
D12
D14
D15
D16
27C512
74LS166
74LS244
74LS374
74LS32
74LS08
74LS37
8-BIT SHIFT REGISTERS
3-state flip flops
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Sheet: /objects/obj_pxl/
File: obj_pxl.kicad_sch
Title: Rally Bike
Size: A3
Date: 2023-05-31
Rev: José Tejada G.
KiCad E.D.A. 8.0.9
Id: 7/12

FINITE STATE MACHINE



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Sheet: /objects/Obj_fms/ File: Obj_fms.kicad_sch			
Title: Rally Bike			
Size: A3	Date: 2023-05-31		**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9			Id: 8/12



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Sheet: /colmix_rgb/
File: colmix_rgb.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 9/12

SHARED BUFFER'S

K1 YM3812-3526 FM OPL
M3 YM3014
M4A M4B LM358
J6 buffers 74LS245
H5 Z80CPU
H3 27128
F1 F2 K3 buffers 74LS245
H1 HM6116P-3 MCM2018 OKI M5126-25 16 BITS SRAM 2KB RAM
K5A K6 decoder 74LS139
L14 address decoder 74LS138
H6A H6B H6C 74LS04
L5A L6A L6B L6D 74LS74 74LS32
74LS244
M1 MB3730 INV. AMP
LS1 Speaker
Jumper_3_Bridged12
R_Network08_US

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File: sound.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 10/12

PWR_FLAG

VCC

CP41 103 CP42 103 CP43 103 CP44 103 CP45 103 CP46 103 CP47 103 CP48 103 CP49 103 CP50 103 CP51 103 CP52 103 CP53 103 CP54 103 CP55 103 CP56 103 CP57 103 CP58 103 CP59 103 CP60 103

CP61 103 CP62 103 CP63 103 CP64 103 CP65 103 CP66 103 CP67 103 CP68 103 CP69 103 CP70 103 CP71 103 CP72 103 CP73 103 CP74 103 CP75 103 CP76 103 CP77 103 CP78 103 CP79 103 CP80 103

CP2 10µF CP3 10µF

74LS08 K16C (10, 9, 8)

R63 470Ω CP90 10nF

J41E 74LS08 | J42E 74LS08 | L41C 74LS74 | L42E 74LS32 | H41G 74LS04 | F41E 74LS32 | E41C 74LS74 | H42C 74LS74 | J43C 74LS74 | D43E 74LS32 | J44G 74LS04 | J45E 74LS08 | H43C 74LS20 | H44E 74LS08 | L43G 74LS14

K44C 74LS139 decoder | L44C 74LS74 | L45E 74LS08

D44G 74LS04 | K45E 74LS32 | K46C 74LS74 | D41E 74LS08 | K47E 74LS08 | D42E 74LS08 | K41E 74LS08 | A41G 74LS04 | K42E 74LS08 | B5C 74LS393 BINARY COUNTER | A5C 74LS393 BINARY COUNTER | A3C 74LS393 BINARY COUNTER | D11C 74LS74 | C5C 74LS74 | D18E 74LS08 | F23E 74LS32 | C6E 74LS08

M41C LM358 (4 V− 8 V+)

E21C 74LS139 decoder

VCC

K43E 74LS38 | D7E 74LS32 | D6E 74LS08 | D1E 74LS08 | D5E 74LS32

GND

PWR_FLAG

VSS

K44B 74LS139 (A1 A0 E | O0 O1 O2 O3) decoder

E21B 74LS139 (A1 A0 E | O0 O1 O2 O3) decoder

B5B 74LS393 (CP MR | Q0 Q1 Q2 Q3) BINARY COUNTER

K18B | C5B | H24B | J22B (D CP PR CLR | Q Q̄) 74LS74

D1A D1B D1C 74LS08

D2C D2D D2E D2F 74LS04

D7B D7C D7D 74LS32

D6A 74LS08 | F24A 74LS32 | K17C 74LS32

D5B 74LS32 | H6A 74LS04

J20B J20D 74LS04

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File: capacitors_lu.kicad_sch

Title: Rally Bike

Size: A3	Date: 2023-05-31	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 11/12

Rally Bike



Rally Bike is a racing arcade video game originally developed by Toaplan and published by Taito in May 1988. In Rally Bike, players compete against computer-controlled opponents in races across locations in the United States. Initially released in arcades, the game was ported to the Nintendo Entertainment System and X68000 by different developers.
Rally Bike was met with mixed reception from video game magazines and dedicated outlets that reviewed the game when it launched in arcades and its ports. As of 2019, the rights to the title is owned by Tatsujin, a company founded in 2017 by former Toaplan member Masahiro Yuge and now-affiliate of Japanese arcade manufacturer exA-Arcadia alongside many other Toaplan IPs.

HARDWARE DESCRIPTION

Main CPU : 68000 @ 10 MHz
Sound CPU : Z80 or Z180 @ 3.5 MHz
Sound Chip : YM3812 @ 3.5 MHz
Other : Demons World has an additional TMS32010 @ 14MHz
BCU : NEC D6508IR077
SCU : NEC D65024GF035

V • T • E	Toaplan	
Games	**1980s**	*Performan* • *Tiger-Heli* • *Guardian* • *Slap Fight* • *Mahjong Sisters* • *Flying Shark* • *Wardner* • *Twin Cobra* • ***Rally Bike*** • *Truxton* • *Hellfire* • *Twin Hawk* • *Demon's World* • *Zero Wing* • *Fire Shark*
	1990s	*Snow Bros.* • *Out Zone* • *Teki Paki* • *Vimana* • *Ghox* • *Pipi & Bibi's* • *Truxton II* • *FixEight* • *Dogyuun* • *Grind Stormer* • *Knuckle Bash* • *Enma Daiō* • *Batsugun* • *Snow Bros. 2: With New Elves*
Related games	*MUSHA* • *Sorcer Striker* • *Twin Cobra II*	
People	Tatsuya Uemura • Junya Inoue • Yusuke Naora	
See also	Tamsoft • Eighting • CAVE • Gazelle • Takumi Corporation	
Category		
Portals: 1980s • Japan • Video games		

Source: https://en.wikipedia.org/wiki/Rally_Bike
http://www.system16.com/hardware.php?id=966&gid=3368#3368

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Sheet: /description/
File: description.kicad_sch
Title: Rally Bike
Size: A3 | Date: 2023-05-31 | **Rev: José Tejada G.**
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